

# **Software Development Fundamentals (SDF) – I**

## **ODD 2023**



### **Lecture 3**

**Jaypee Institute of Information Technology (JIIT), Noida**

# **Outline of Lecture**

- 1. Puzzle Set 2 (Three Fastest Horses) – Design the Logic**
- 2. Puzzle Set 2 (Fake Coin) – Design the Logic**
- 3. Exercises**

## Puzzle Set 2: Three Fastest Horses

**Problem Statement:** It is desired to find out the Fastest, 2<sup>nd</sup> Fastest, and 3<sup>rd</sup> Fastest horses out of given 36 horses. There is a race course in which you can conduct the race for horses, but at a time maximum 6 horses can participate in the race. You don't have the access to Clock/Stopwatch, *i.e.* you can not find the actual speed of the horse in the race, however, you can observe/find out the relative speed of the 6 horses participated in a race. Minimum how many races you must have to conduct to find out the three fastest horses out of 36 horses.

**Given**  
36 Horses



**Desired**  
Three Fastest Horses



**Constraint: 1**  
Maximum 6 Horses in one Race



**Constraint: 2**  
No Clock/Stopwatch



<https://us.123rf.com/450wm/pteshka/pteshka130351110500414/40256659-stock-vector-horse-race-derby-equestrian-sport-silhouette-of-racing-horse-with-jockey-on-isolated-background-hors.jpg?ver=6>

**Fastest**

7493.jpg

<https://cdn5.vectorstock.com/i/1000x1000/38/64/stopwatch-vector-913864.jpg>

## Puzzle Set 2: Three Fastest Horses

### DESIGN the LOGIC (Logic 1)

|        | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | 4 <sup>th</sup> | 5 <sup>th</sup> | 6 <sup>th</sup> |
|--------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Race 1 | H1              | H2              | H3              | H4              | H5              | H6              |
| Race 2 | H7              | H8              | H9              | H10             | H11             | H12             |
| Race 3 | H13             | H14             | H15             | H16             | H17             | H18             |
| Race 4 | H19             | H20             | H21             | H22             | H23             | H24             |
| Race 5 | H25             | H26             | H27             | H28             | H29             | H30             |
| Race 6 | H31             | H32             | H33             | H34             | H35             | H36             |

|        | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | 4 <sup>th</sup> | 5 <sup>th</sup> | 6 <sup>th</sup> |
|--------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Race 7 | H1              | H7              | H13             | H19             | H25             | H31             |

### Answers obtained from Logic 1

Races: 7

Fastest: H1

2<sup>nd</sup> Fastest: H7

3<sup>rd</sup> Fastest: H13

**Q1.** Are, H1, H7, and H13 the three fastest horses?

**Ans.** No

**Argument.** H1 is fastest of Race 1 and fastest of Race 7 (*i.e.* race of fastest horses of each race)

Hence, H1 is the fastest horse out of 36.

Do we know, which one is faster, H2 or H7/H13?

Without comparing the relative speed of H2 and H7/H13, should we claim H7 is faster

## Puzzle Set 2: Three Fastest Horses

### DESIGN the LOGIC (Logic 2)

|        | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | 4 <sup>th</sup> | 5 <sup>th</sup> | 6 <sup>th</sup> |
|--------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Race 1 | H1              | H2              | H3              | H4              | H5              | H6              |
| Race 2 | H7              | H8              | H9              | H10             | H11             | H12             |
| Race 3 | H13             | H14             | H15             | H16             | H17             | H18             |
| Race 4 | H19             | H20             | H21             | H22             | H23             | H24             |
| Race 5 | H25             | H26             | H27             | H28             | H29             | H30             |
| Race 6 | H31             | H32             | H33             | H34             | H35             | H36             |

|         | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | 4 <sup>th</sup> | 5 <sup>th</sup> | 6 <sup>th</sup> |
|---------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Race 7  | H1              | H2              | H3              | H7              | H8              | H9              |
| Race 8  | H13             | H14             | H15             | H19             | H20             | H21             |
| Race 9  | H25             | H26             | H27             | H31             | H32             | H33             |
| Race 10 | H1              | H2              | H3              | H13             | H14             | H15             |
| Race 11 | H1              | H2              | H3              | H25             | H26             | H27             |

Answers obtained from Logic 2

Races: 11; Fastest: H1; 2<sup>nd</sup> Fastest: H2; 3<sup>rd</sup> Fastest: H3

**Q1.** Are, H1, H2, and H3 the three fastest horses?

**Ans.** Yes

**Argument.** We compared H1, H2, and H3 with all possible contenders to decide them as three fastest horses

**Q2.** Is 11 the minimum number of races we must conduct to find three fastest horses out of 36 horses?

**Ans.** Can not say? Lets explore other logic too

# Puzzle Set 2: Three Fastest Horses

## DESIGN the LOGIC (Logic 3)

|        | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | 4 <sup>th</sup> | 5 <sup>th</sup> | 6 <sup>th</sup> |
|--------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Race 1 | H1              | H2              | H3              | H4              | H5              | H6              |
| Race 2 | H7              | H8              | H9              | H10             | H11             | H12             |
| Race 3 | H13             | H14             | H15             | H16             | H17             | H18             |
| Race 4 | H19             | H20             | H21             | H22             | H23             | H24             |
| Race 5 | H25             | H26             | H27             | H28             | H29             | H30             |
| Race 6 | H31             | H32             | H33             | H34             | H35             | H36             |

|        | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | 4 <sup>th</sup> | 5 <sup>th</sup> | 6 <sup>th</sup> |
|--------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Race 7 | H1              | H7              | H13             | H19             | H25             | H31             |

In Logic 1, we found H1 as the fastest horse, however, we can not tell about H7 & H13, as other contenders were not explored for remaining positions

**Q1.** Besides H1, who are the contenders for remaining positions?

**Ans.** H7, H13, H2, H3, and H8

**Argument 1.** H13 is the fastest in Race 3 & 3<sup>rd</sup> fastest in Race 7  
*i.e.* only H13 of Race 3 can be the contenders

**Argument 2.** H7 is the fastest in Race 2 & 2<sup>nd</sup> fastest in Race 7  
*i.e.* besides H7, H8 of Race 2 can be the contenders

**Argument 3.** H1 is the fastest in Race 1 & fastest in Race 1  
*i.e.* H2 and H3 of Race 1 can also be the contenders

So, one more race between possible contenders for second and third fastest horses out of 36

|        | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | 4 <sup>th</sup> | 5 <sup>th</sup> | 6 <sup>th</sup> |
|--------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Race 8 | H1              | H2              | H3              | H7              | H8              | H13             |

## Puzzle Set 2: Three Fastest Horses

|        |     |                |                |                |                |                |
|--------|-----|----------------|----------------|----------------|----------------|----------------|
| Race 3 | H13 | <del>H14</del> | <del>H15</del> | <del>H16</del> | <del>H17</del> | <del>H18</del> |
|        |     |                |                |                |                |                |
| Race 2 | H7  | H8             | <del>H9</del>  | <del>H10</del> | <del>H11</del> | <del>H12</del> |
|        |     |                |                |                |                |                |
| Race 1 | H1  | H2             | H3             | <del>H4</del>  | <del>H5</del>  | <del>H6</del>  |

## Puzzle Set 2: Fake Coin

**Problem Statement:** You have been given 8 identical looking coins. Out of 8, seven coins are genuine/real coins whereas one coin is fake, but as they look identical so you can not differentiate. However, you know that the fake coin is lighter than a genuine coin. You have a two pan weighing scale, but without weights, i.e. you cannot measure the weights, however coins can be compared for equal weights or unequal weights (lighter/heavier). It is desired to find out the minimum number of weighing needed to identify the fake coin.

### Given:

8 identical coins, 7 genuine, 1 fake

### Desired:

Find fake coin

### Available:

Two pan weighing scale

### Constraint:

Weights are not available

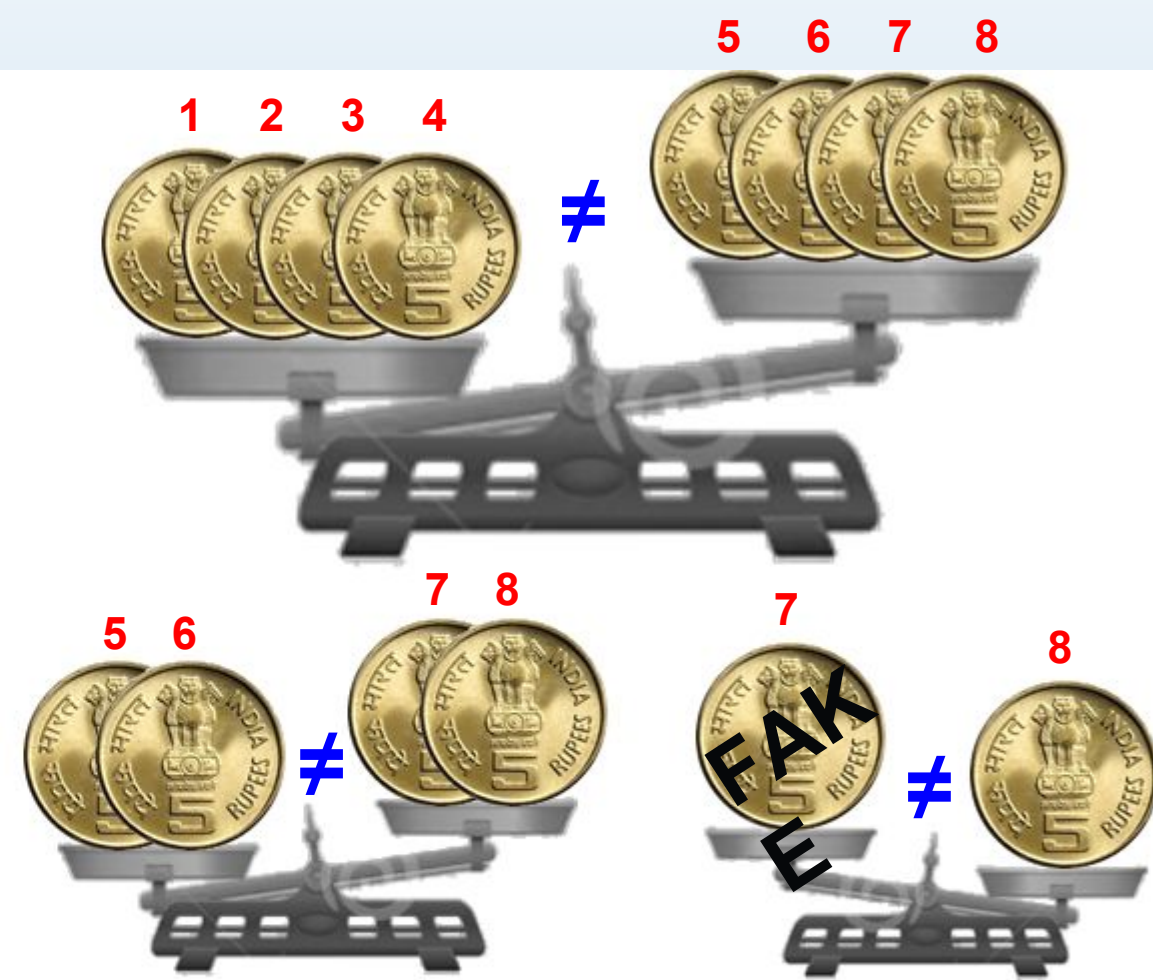


# Puzzle Set 2: Fake Coin

## DESIGN the LOGIC



**Q1.** Did we find the fake coin. How many weighing  
**Ans.** Yes. It is Coin No 7. **Total 4 weighing**  
**Q2.** Is it the Minimum number of required weighing  
**Ans.** Cannot say. Let's think and Design better



**Q1.** Did we find the fake coin. How many weighing  
**Ans.** Yes. It is Coin No 7. **Total 3 weighing**  
**Q2.** Is the applied number of weighing Minimum  
**Ans.** Cannot say. Let's think and Design better

# Puzzle Set 2: Fake Coin

DESIGN the LOGIC (Logic 3)

Scenario 1



Scenario 2



What we infer?

EITHER

EITHER

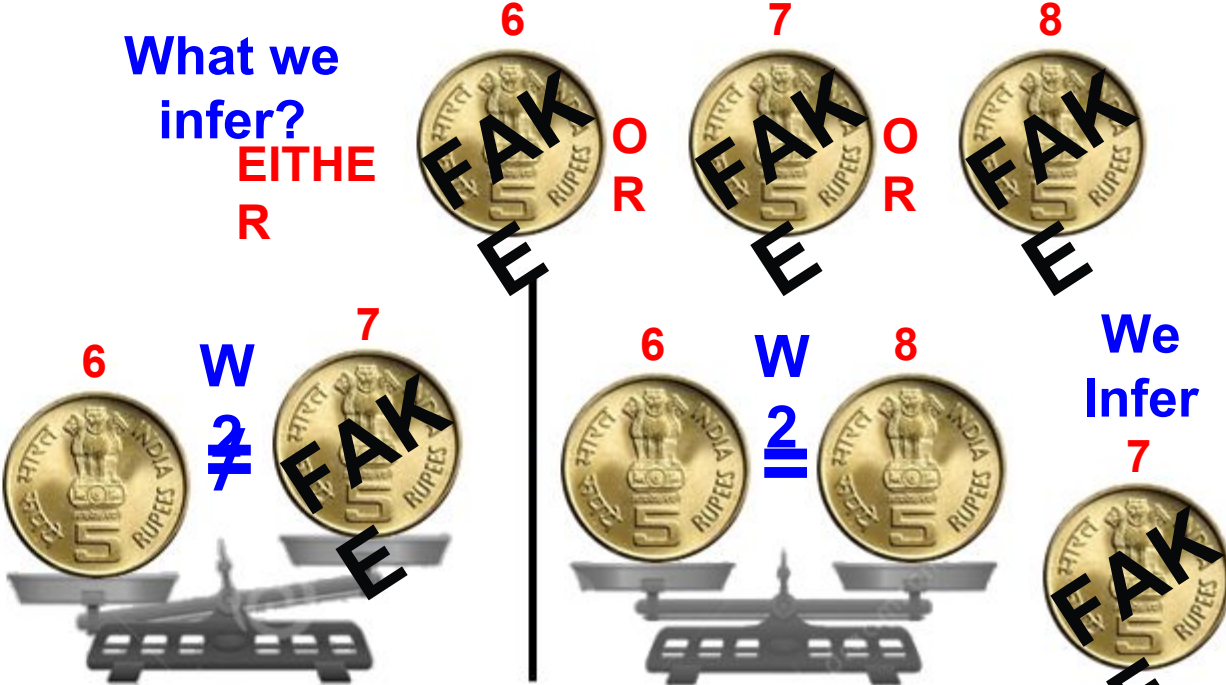
OR

OR

OR

We Infer

7



# Exercises

- Q1.** Find out the minimum count of races required to identify three fastest horses, if count of horses is 25 and at a time only 5 horses can participate in a race. Also, you do not have the accessibility to watches.
- Q2.** Let us modify the Fake Coin puzzle as follows: 8 coins are given, out of which 1 coin is fake; you do not know whether the fake coin is lighter or heavier than the genuine coin. You have accessibility of two pan weighing machine without weights. Identify the minimum count of required weighing to identify the fake coin.

**End of  
Lecture 3**